

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO3_AT2 — DATA ANALYSIS PROJECT

Course Code:	DSA0606	Course Title:	Data Handling and Visualization
CO Assessed:	CO3 – Analysis (BL4)	Assessment:	Assessment Tool 2 – Data Analysis Project
Weightage:	40% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO3 Statement:	<i>Analyze real datasets using ECDFs, skewed-distribution techniques, Q–Q plots, multi-distribution comparisons, and proportion visualizations to identify patterns and justify conclusions.(BL4)</i>		
Student Name:	P.Vishwa nandan	Reg. No.:	192473015
Section / Batch:	2024	Date:	13-08-2026

Code of Conduct

/ P.Vishwa Nandan (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: vishwanandan

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must go beyond construction: analyze the given data, identify the relevant pattern, and justify your conclusion.
- Show all supporting calculations (e.g., IQR, percentiles, angle calculations) where relevant.
- Where a question asks you to compare or recommend, state your reasoning explicitly — a correct chart choice without justification will not receive full marks.

Annexure A – DATA ANALYSIS PROJECT

Rubrics for Calculation ASSESSMENT RUBRIC
AT2 — Data Analysis Project
CO3 · BT4: Analyze ·

This rubric is used to assess student responses to the CO3-AT2 Data Analysis Project, in which students analyze a given dataset using ECDF, skewed-distribution, Q–Q plot, multi-distribution, or proportion-visualization techniques and justify their conclusions.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of the relevant technique (ECDF, skewed-distribution/log transform, Q–Q plot, multi-distribution comparison, or proportion visualization) and correctly identifies which pattern applies to the given data.	Good understanding of the relevant technique, with only minor gaps in identifying the correct pattern.	Basic understanding, but with some misconceptions about which technique or pattern applies to the data.	Poor understanding of core concepts; misidentifies the pattern or technique required by the data.
Explanation	25%	Provides detailed, well-reasoned analysis explicitly tied to the specific values in the dataset, going beyond description to explain why the pattern occurs or what it implies.	Provides clear analysis with adequate reasoning tied to the data, for most parts.	Analysis is present but generic or only loosely connected to the specific dataset given.	Little or no analytical reasoning provided; answers restate the data without interpretation.

Application	20%	Effectively applies analytical techniques to compare, justify, or recommend a conclusion that is well-supported by the data and any calculations shown.	Applies analytical techniques correctly, with only minor errors in reasoning or calculation.	Limited application; conclusions are only partially supported by the data or calculations.	Fails to apply appropriate analytical techniques; conclusions are unsupported or incorrect.
Organization	15%	The response is clearly structured, addressing every sub-part (a–b) in a logical sequence with coherent overall flow.	The response is organized and addresses each sub-part, with only minor issues in flow or clarity.	Basic structure; sub-parts are addressed but the response lacks clarity or a logical sequence.	Poorly organized; the response is incoherent, and sub-parts are left unaddressed or answered out of order.
Accuracy	10%	All parts of the answer — pattern identification, calculations, and conclusions — are completely accurate and well-suited to the data.	Mostly accurate, with only minor omissions or a small error in one sub-part.	Partially correct; some elements are missing, mismatched to the data, or incorrect.	Largely incorrect or inappropriate, with major gaps across multiple sub-parts.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Problem 2

Code:

```
# Daily sales data
storeA <- c(40,42,45,46,48,50,52,53,55,56,
           58,60,61,63,65,66,68,70,72,75)

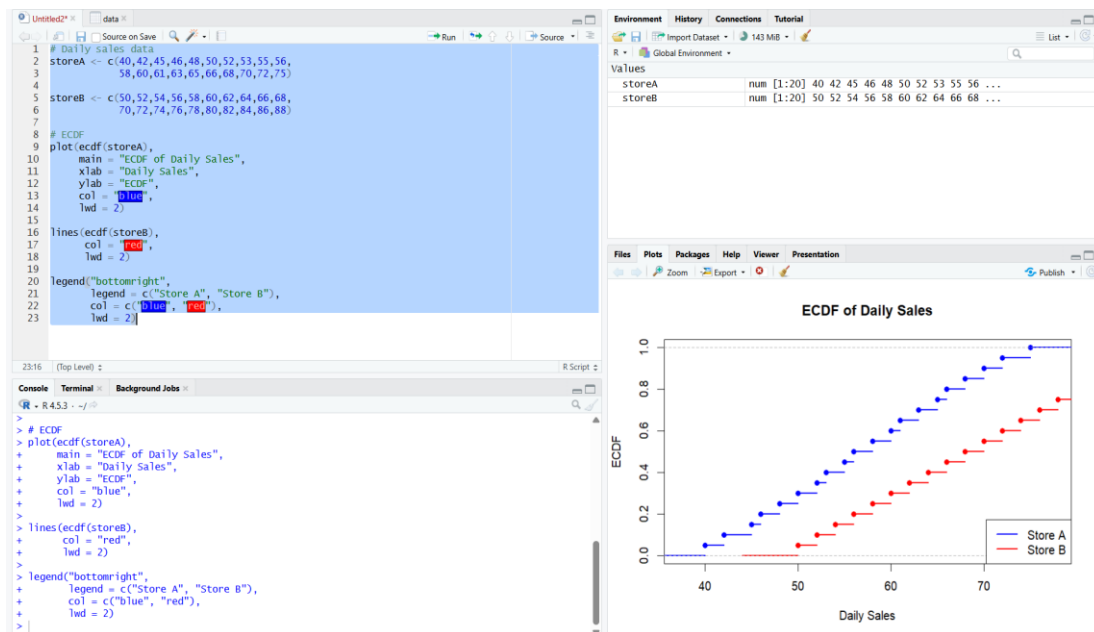
storeB <- c(50,52,54,56,58,60,62,64,66,68,
           70,72,74,76,78,80,82,84,86,88)

# ECDF
plot(ecdf(storeA),
     main = "ECDF of Daily Sales",
     xlab = "Daily Sales",
     ylab = "ECDF",
     col = "blue",
     lwd = 2)

lines(ecdf(storeB),
      col = "red",
      lwd = 2)

legend("bottomright",
      legend = c("Store A", "Store B"),
      col = c("blue", "red"),
      lwd = 2)
```

Output:



Problem 8

Code:

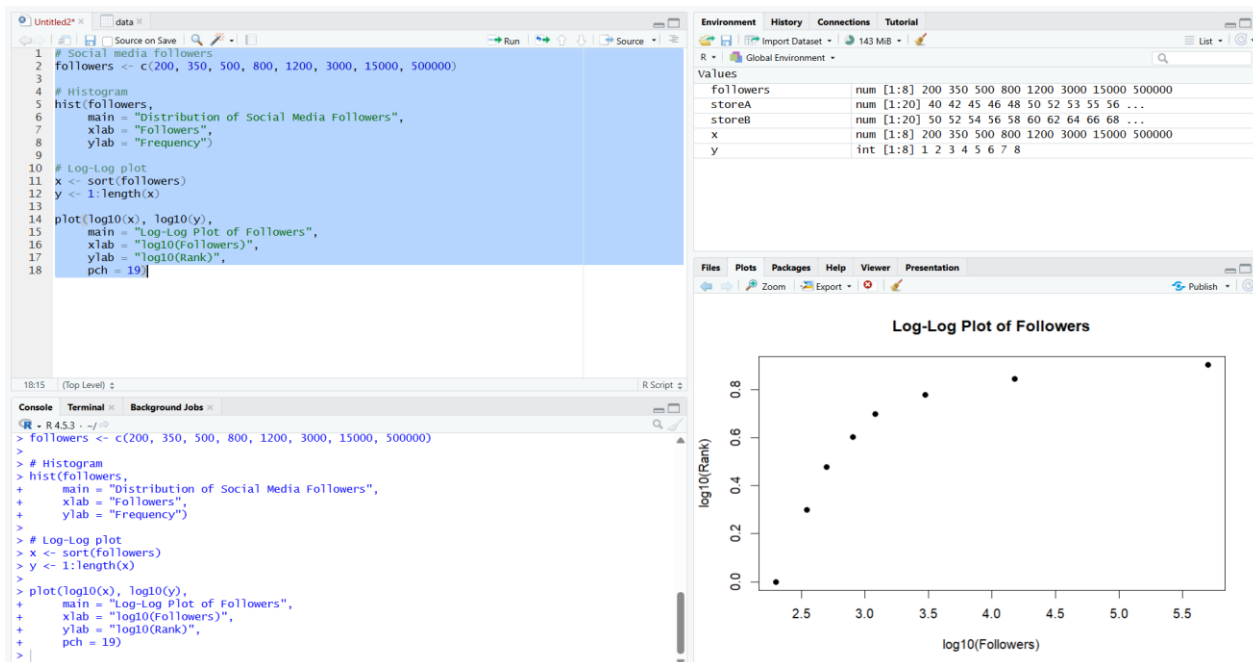
```
# Social media followers
followers <- c(200, 350, 500, 800, 1200, 3000, 15000, 500000)

# Histogram
hist(followers,
     main = "Distribution of Social Media Followers",
     xlab = "Followers",
     ylab = "Frequency")

# Log-Log plot
x <- sort(followers)
y <- 1:length(x)

plot(log10(x), log10(y),
     main = "Log-Log Plot of Followers",
     xlab = "log10(Followers)",
     ylab = "log10(Rank)",
     pch = 19)
```

Output:



Problem 14

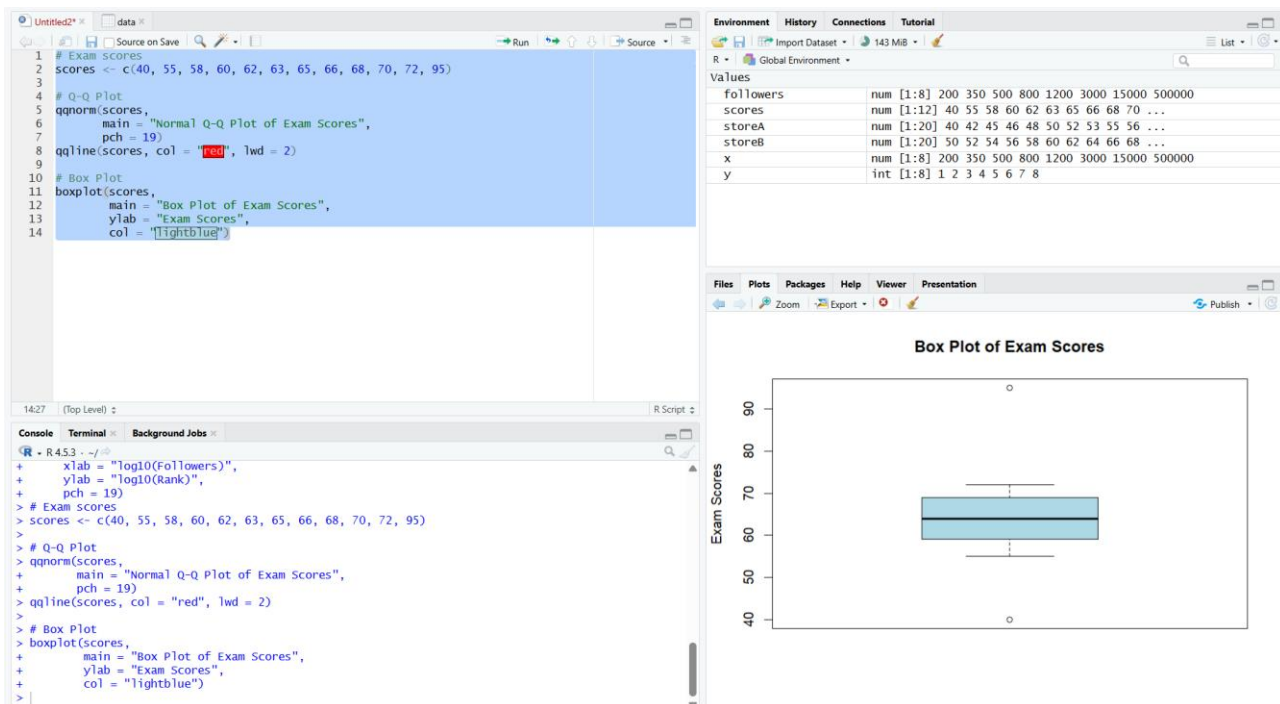
Code:

```
# Exam scores
scores <- c(40, 55, 58, 60, 62, 63, 65, 66, 68, 70, 72, 95)
```

```
# Q-Q Plot
qqnorm(scores,
  main = "Normal Q-Q Plot of Exam Scores",
  pch = 19)
qqline(scores, col = "red", lwd = 2)
```

```
# Box Plot
boxplot(scores,
  main = "Box Plot of Exam Scores",
  ylab = "Exam Scores",
  col = "lightblue")
```

Output:



Problem 20

Code:

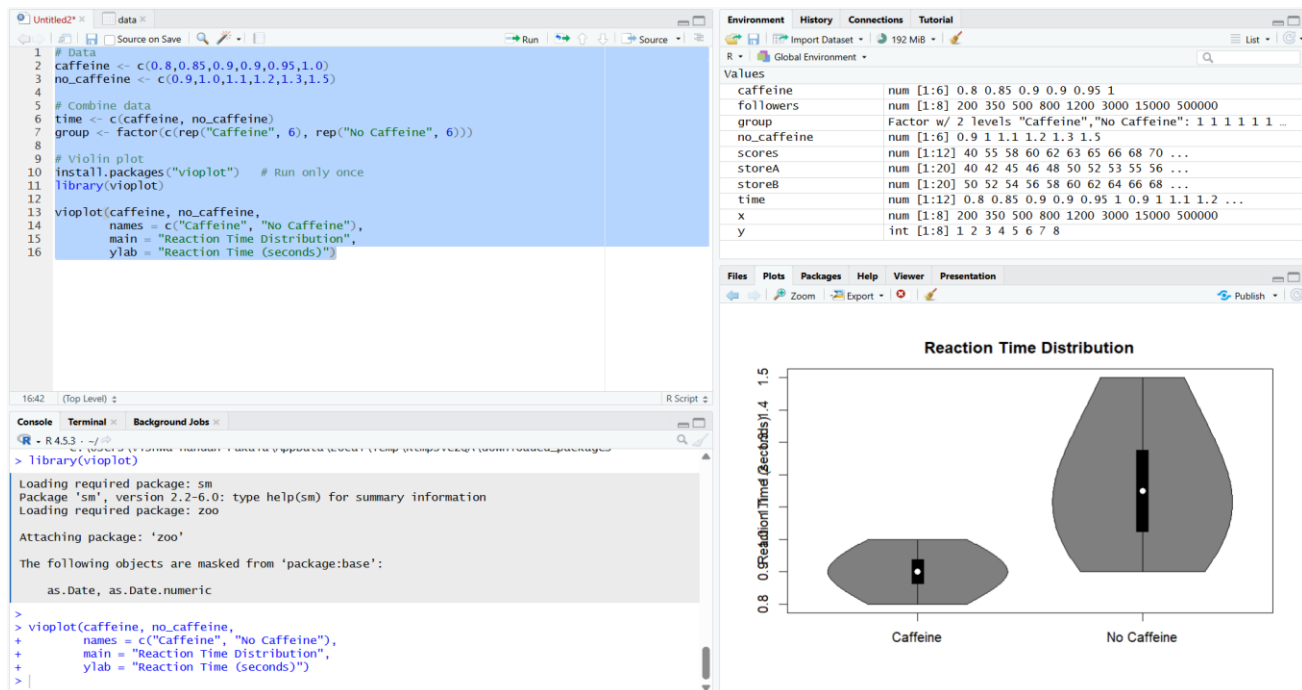
```
# Data
caffeine <- c(0.8,0.85,0.9,0.9,0.95,1.0)
no_caffeine <- c(0.9,1.0,1.1,1.2,1.3,1.5)

# Combine data
time <- c(caffeine, no_caffeine)
group <- factor(c(rep("Caffeine", 6), rep("No Caffeine", 6)))

# Violin plot
install.packages("vioplot") # Run only once
library(vioplot)

vioplot(caffeine, no_caffeine,
        names = c("Caffeine", "No Caffeine"),
        main = "Reaction Time Distribution",
        ylab = "Reaction Time (seconds)")
```

Output:



Problem 21

Code:

```
# Data
sales <- c(200000, 50000)
names(sales) <- c("Store A", "Store B")

# Sales composition
data <- matrix(c(50,30,20,
                 50,30,20),
              nrow = 2, byrow = TRUE)

rownames(data) <- c("Store A", "Store B")
colnames(data) <- c("Electronics", "Clothing", "Other")

# 100% stacked bar chart
barplot(t(data),
        beside = FALSE,
        col = c("skyblue", "orange", "lightgreen"),
        legend = TRUE,
        main = "Sales Composition (%)",
        ylab = "Percentage")
```

Output:

